

DemoGen: A Framework for Auditing and Mitigating Demographic Bias in LLM-Generated Content

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Abstract

LLMs are widely used for high-stakes writing tasks – cover letters, recommendation letters, and professional advice – where output quality directly affects the individuals named in prompts. We present **DemoGen**, a framework that audits and mitigates demographic bias in LLM-generated content by pairing racially associated names with task templates, generating outputs under four conditions (baseline, persona-blind, self-consistency reranking with $k = 3$ –5 candidates, and a system-prompt intervention), and measuring parity gaps across LLM-as-judge quality scores, response length, and readability.

Across 540 prompts spanning five task domains and three perceived racial groups (White, Black, Asian), baseline outputs show statistically significant quality disparities (one-way ANOVA $p = 0.040$, max parity gap = 0.110). The gap is driven by a single contrast: Asian-associated versus Black-associated names ($p = 0.016$, Cohen’s $d = 0.255$, Bonferroni threshold $p < 0.0167$). Persona-blind prompting reduces the maximum quality gap to 0.017 (85% reduction) and raises average quality from 4.74 to 4.98 under the Llama judge; this quality increase did not replicate under an independent Mistral judge. Reranking achieves partial improvement (gap = 0.063) but is limited because all candidates are drawn from already-biased prompts. A system-prompt instruction to ignore demographics reduces the gap to 0.072 but does not match persona-blind, pointing to conditioning that operates below the level of explicit instruction following. Disparities concentrate in cover-letter writing (baseline gap = 0.269); parity gaps evaluated only at the global mean level will miss this. Name-based demographic disparity appears largely driven by input cues and can be substantially reduced through prompt-design changes without retraining the model.

1 Introduction

LLMs are already used for cover letters, recommendations, and explanations where output quality affects real outcomes for the named individual. Unlike classification bias, this form is not explicitly discriminatory – the model may produce more polished or detailed text for some names than others, and neither user nor developer would necessarily notice. Bertrand and Mullainathan [1] showed that racially associated names elicit differential treatment in hiring; if LLMs encode analogous associations from training data, they may produce measurably different outputs for prompts that differ only in the name – a form of *demographic conditioning* that is invisible by design.

We ask:

Do LLMs produce systematically different quality outputs across prompts grouped by perceived race-associated names when the only variation is the name used, and can simple prompting interventions eliminate this disparity?

We address this through DemoGen, which generates responses to the same templates using names associated with White, Black, and Asian groups, and compares four conditions: baseline, persona-blind, reranked, and system-prompt. We expected differences to show up more in surface features like length than in LLM-as-judge quality scores. The quality gap concentrated in cover letters rather than across tasks, and persona-blind prompting outperformed reranking by more than expected.

Our main contributions:

1. A reproducible prompt suite of 540 balanced prompts across 5 task types and 3 perceived racial groups, drawn from established demographic name lists.
2. An audit pipeline computing LLM-as-judge quality scores, response length, and Flesch-Kincaid readability, with pairwise parity gaps, one-way ANOVA, and Bonferroni-corrected t -tests.
3. An empirical comparison of four prompting conditions: baseline (names present), persona-blind (names removed), reranking ($k = 3-5$), and a system-prompt intervention.
4. Evidence that persona-blind reduces the maximum quality gap by 85%, that disparities concentrate in cover-letter writing, and that system-prompt instructions produce only partial mitigation.

Two hypotheses drive the evaluation: **H1** – baseline outputs show statistically significant quality disparities across prompts grouped by perceived race-associated names; **H2** – persona-blind reduces those disparities more than output-level or instruction-level interventions. Both are evaluated in Section 5.

Name-based audit studies. [1] sent resumes with stereotypically White or Black names to real job postings and found a 50% gap in callback rates. We use their name lists directly for comparability with a standard audit reference, applying the same logic: vary only the name, hold everything else constant.

Bias in language representations. [3] showed that word embeddings trained on large corpora encode human-like social biases, including racial and gender associations. A name is not just a label – it is a signal that can activate learned associations inside the model, which is why we vary names while keeping task content constant.

Fairness metrics for NLP. [2] formalize unintended bias metrics for text classifiers using parity gaps – absolute differences in performance across demographic groups. We adopt this metric for its simplicity and interpretability. [4] survey bias evaluation and mitigation for LLMs, confirming that generation-time bias is less studied than classification-time bias.

Benchmarks for social bias in generation. Prior work shows that demographic bias can appear in long-form LLM-generated professional writing, not only in short classification benchmarks. [7] benchmark language agency bias in generated reference letters, reporting race-related language-agency bias scores of 8.02 for ChatGPT, 7.90 for Mistral, and 26.29 for Llama3, with overall reference-letter bias scores of 27.91, 37.53, and 40.39 respectively. [6] similarly find significant gender-linked differences in LLM-generated recommendation letters, including agency differences

($t = 10.47$) and positivity differences ($t = 5.93$, $p = 1.58 \times 10^{-9}$). [5] introduce SAGED, a holistic pipeline for social-bias benchmarking with customizable fairness calibration.

These studies use agency and style metrics; DemoGen uses LLM-as-judge quality parity gaps, so numeric scales are not directly comparable. [7] report a Llama3 reference-letter agency bias score of 26.29 – among the highest across models they tested. Our LABE-inspired agency proxy on Llama 3.1 8B found a non-significant race-linked agency gap (ANOVA $p = 0.925$), suggesting the newer model generation or our broader task mix (dominated by cover letters rather than reference letters) may attenuate agency-level bias even when quality-level bias is detectable. [6] found significant gender gaps in LLM-generated recommendation letters (agency $t = 10.47$); our supplementary gender check on cover-letter writing found no significant quality gap under the Llama judge, consistent with race-associated name effects being the dominant signal in our prompt suite, and with gender effects being more visible in agency-style metrics than in quality scores. All three papers confirm that demographic cues affect long-form generated professional text – the same setting we target.

We built our own prompt suite rather than adopting existing benchmarks because we wanted practical long-form tasks where output quality has real consequences. DemoGen extends this line of work across five writing tasks and four mitigation strategies: baseline gap 0.110; persona-blind reduced it to 0.017; reranking to 0.063; system prompting to 0.072. Unlike decoding-level interventions [7] or multi-judge pipelines like SAGED [5], persona-blind prompting requires no model access, no decoding modifications, and no human annotation – applicable to any black-box API at zero additional inference cost.

2 Framework Overview

DemoGen is implemented as a modular Python pipeline with four sequential stages, as illustrated in Figure 1.



Figure 1: DemoGen four-stage pipeline. The prompt suite drives generation under four conditions; the audit pipeline computes fairness metrics; the visualization stage produces figures and summary reports.

The repository is organized into `src/generate`, `src/audit`, and `src/visualize` sub-packages. All experiment outputs are stored in `results/` as `.csv` files and `.json` response logs, enabling full reproducibility. Each response JSON is keyed by `prompt_id`, which links it back to the original prompt suite row, the demographic name used, the task type, and the perceived race label, so any result in the audit CSVs can be traced to the exact prompt that produced it.

3 Methodology

3.1 Prompt Suite Construction

Task domains. We define five task types relevant to professional and educational contexts: **cover letters**, **recommendation letters**, **concept explanations**, **advice giving**, and **problem solving**. Each task type has 20 distinct template prompts, yielding 100 unique templates in total.

Demographic name lists. We use racially associated first names drawn from [1] and [3] for three perceived racial groups: **White** (e.g., Todd, Emily, Sarah), **Black** (e.g., Jamal, Lakisha, Ebony), and **Asian** (e.g., Arjun, Hiroshi, Wei). Names are substituted into the [NAME] slot in each template, creating demographically differentiated prompt instances. The full prompt suite contains $5 \times 3 \times 36 = 540$ prompts, balanced across task types and racial groups. Here, 36 refers to prompt instances per task-race cell, not 36 unique names. The full suite uses 37 unique names total: 11 Asian-associated, 12 Black-associated, and 14 White-associated names. Names repeat across different prompt contexts, so each task-race group contains 36 prompt rows spread across multiple names. The 36 rows per cell are drawn without replacement from the full template $\times$ name cross-product, so a given template may appear up to four times within one cell paired with different names. Responses sharing the same template are therefore not fully independent; this template clustering is a limitation on strict independence assumptions.

Experimental design. The prompt suite samples balanced prompt instances across task templates and race-associated names so that each task-race cell contains 36 rows. Because within-race name variance exceeded between-race variance in the baseline condition, we interpret results as a name-based demographic cue audit rather than a pure estimate of racial group effects. An example pair of matched prompts is shown below.

Baseline (White): “Write a cover letter for **Emily** applying for a software engineer position...”

Baseline (Black): “Write a cover letter for **Lakisha** applying for a software engineer position...”

3.2 Generation Methods

We compare four conditions, summarized in Table 1.

Table 1: Summary of the four generation conditions used in DemoGen. *Note for RERANKED: selecting by proximity to baseline mean anchors to a biased reference; see Section 6.2.*

Condition	Description
BASILINE	Name included as written; model receives the full demographic signal. Temperature $T = 0$ for determinism. Represents the current default behavior of most deployed LLM applications.
PERSONA-BLIND	All demographic names replaced with “the applicant” before generation, removing the primary demographic cue while preserving task framing. Temperature $T = 0$.
RERANKED	$k = 3$ –5 candidates per prompt (mixed- k ; see Section 4). Each scored by an LLM-as-judge; the candidate whose score is closest to the per-task-type mean quality established by the baseline is selected. $T = 0.7$.
SYSTEM-PROMPT	Demographic names retained in user prompt; a system-level instruction added before generation directing the model to disregard demographic information. $T = 0$. Evaluated on the full 540-prompt suite.

3.3 Evaluation Metrics

For each response we compute the following three metrics.

- **LLM-as-judge quality** ($q \in [1, 5]$): A prompted LLM evaluates each response on a rubric covering accuracy, relevance, clarity, and completeness, and returns an integer score. This is the primary fairness-relevant metric. *Methodological note:* the same model (Llama 3.1 8B Instant) is used for both generation and quality scoring. This creates a potential self-consistency bias where the judge may systematically rate its own outputs higher for certain demographic groups. We partially address this in Section 6.6 via an independent Mistral judge, but readers should interpret absolute quality scores with this confound in mind.
- **Response length** (word count): Systematic length differences across groups can indicate disparate treatment even when quality scores are equal.
- **Readability** (Flesch-Kincaid Grade Level): Systematic differences in readability could reflect different implicit assumptions about the reader’s background based on their perceived demographic group. We include readability as a secondary fairness-relevant signal because differential reading-level targeting across demographic groups would indicate implicit assumptions about audience background, even when quality scores are equal.

Parity gap. For each metric m and pair of demographic groups g_1, g_2 , we define the parity gap as:

$$\Delta_{g_1, g_2}(m) = |\mathbb{E}[m \mid g_1] - \mathbb{E}[m \mid g_2]|$$

We report the *maximum pairwise gap* across all group pairs as the primary summary statistic, following [2].

Statistical tests. We run one-way ANOVA across the three racial groups for each metric within each condition. For significant ANOVA results we follow up with Welch pairwise t -tests using Bonferroni correction, as Welch’s test does not assume equal variances across demographic groups. We also report Cohen’s d effect sizes to characterize practical significance independent of sample size.

4 Experiments

4.1 Experimental Setup

All four conditions produce 540 selected responses each, yielding $540 \times 4 = 2,160$ total selected responses. The RERANKED condition used a mixed candidate budget due to API rate-limit constraints during data collection: 205 prompts were run with $k = 5$ and 335 with $k = 3$, yielding 2,030 total candidate generations (verified by counting the `k` field across all reranked response JSON files), from which the final response per prompt is selected via LLM-as-judge scoring.

Table 2 summarizes the full reproducibility configuration for all conditions.

Table 2: Reproducibility configuration for all experimental conditions.

Condition	Model	API	Temp.	k	Final n
BASELINE	Llama 3.1 8B Instant	Groq	0.0	1	540
PERSONA-BLIND	Llama 3.1 8B Instant	Groq	0.0	1	540
RERANKED	Llama 3.1 8B Instant	Groq	0.7	3–5 (mixed)	514*
SYSTEM-PROMPT	Llama 3.1 8B Instant	Groq	0.0	1	540

*26 reranked responses returned non-numeric judge scores and were excluded; quality analysis uses 514 valid entries. Fallback provider: GPT-3.5-turbo (OpenAI). No fallback was triggered during the run.

All responses are generated using **Llama 3.1 8B Instant** (Meta), accessed via the Groq API (`llama-3.1-8b-instant`), with GPT-3.5-turbo (OpenAI) configured as a fallback provider. In practice, all experimental responses were produced by Llama 3.1 8B Instant with no fallback required. The LLM-as-judge scoring used to evaluate response quality and to select the final response in the RERANKED condition is performed by the same model. BASELINE and PERSONA-BLIND generations use $T = 0$ for deterministic outputs; RERANKED candidates use $T = 0.7$ to introduce diversity across the candidate pool. Condition-level quality comparisons were verified to be directionally robust to the reranked sample-size difference; the reported gap of 0.063 is computed on 514 valid responses, and the direction of all comparisons holds when the 26 excluded prompts are assigned the per-task mean score as a conservative imputation (imputed gap: 0.083).

5 Results

5.1 Overall Parity Gaps Across Conditions

Table 3 summarizes the maximum pairwise parity gap for each metric across all four conditions.

Table 3: Maximum pairwise racial parity gaps per metric and condition. Lower values indicate greater fairness across groups.

Condition	LLM Quality	Length (words)	Readability (FKGL)
BASELINE	0.110	5.422	0.767
PERSONA-BLIND	0.017	7.644	0.416
RERANKED	0.063	7.856	0.462
SYSTEM-PROMPT	0.072	5.222	0.771

The BASELINE condition shows the largest quality gap (0.110). SYSTEM-PROMPT has a readability gap (0.771) nearly identical to BASELINE (0.767), while both are substantially larger than PERSONA-BLIND (0.416), indicating that neither condition with names present adequately addresses surface-feature conditioning. PERSONA-BLIND prompting reduces the quality gap to 0.017, an **85% reduction**, while readability also improves substantially. RERANKED generation achieves an intermediate quality gap of 0.063, though readability and length gaps are comparable to PERSONA-BLIND rather than BASELINE.

Response length gaps are larger in PERSONA-BLIND and RERANKED than in BASELINE. When names are absent the model varies output length differently across tasks, and the reranking objective (proximity to task-mean quality) does not penalize length disparity directly.

5.2 Mean Quality and Fairness-Quality Tradeoff

Table 4 reports the mean LLM-as-judge quality score per racial group and overall for each condition.

Table 4: Mean LLM-as-judge quality score by perceived race and condition. RERANKED quality is computed over 514 valid responses (26 non-numeric judge scores excluded).

Condition	Asian	Black	White	Overall
BASELINE	4.805	4.694	4.713	4.737
PERSONA-BLIND	4.972	4.989	4.983	4.981
RERANKED	4.847	4.784	4.815	4.815
SYSTEM-PROMPT	4.867	4.794	4.833	4.831

PERSONA-BLIND achieves the highest mean quality overall (4.981) – higher than BASELINE (4.737) – meaning name removal improves fairness and raises average quality, pointing to demographic conditioning actively degrading generation for some groups. Under BASELINE, Black-named prompts score lowest (4.694), White next (4.713), and Asian highest (4.805) – White-named prompts scored lower than Asian-named, which prior audit studies would not predict. The largest baseline gap (0.110) is between Asian and Black groups. RERANKED achieves intermediate quality (4.815), below PERSONA-BLIND but above BASELINE.

Figure 2 plots the fairness-quality tradeoff directly.

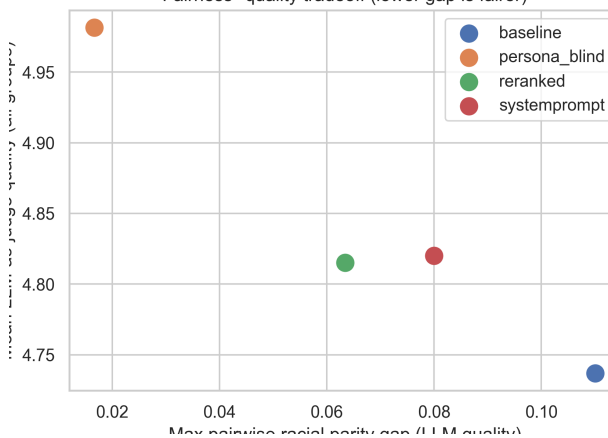


Figure 2: Fairness and quality tradeoff across all four conditions. The horizontal axis shows the maximum pairwise racial parity gap (lower is fairer); the vertical axis shows mean LLM-as-judge quality across all groups. PERSONA-BLIND achieves better fairness and higher measured quality simultaneously, Pareto-dominating both other main conditions. Pareto dominance here means PERSONA-BLIND is strictly better than all other conditions on both axes simultaneously – no other condition achieves a lower gap without also achieving lower quality, and no other condition achieves higher quality without also incurring a larger gap. SYSTEM-PROMPT achieves partial mitigation (gap = 0.072) but remains weaker than PERSONA-BLIND.

PERSONA-BLIND achieves better fairness and higher measured quality simultaneously: it is the fairest (gap = 0.017) *and* the highest quality (mean = 4.981). BASELINE achieves the worst fairness (gap = 0.110) and the lowest mean quality (4.737). RERANKED sits between the two (gap = 0.063, mean = 4.815).

Note that the average quality improvement under PERSONA-BLIND is judge-sensitive: an independent Mistral judge did not replicate this increase (Section 6.6), so the quality claim should be interpreted as a Llama-judge result rather than a model-independent finding.

5.3 Statistical Tests

Table 5 reports one-way ANOVA results and Bonferroni-corrected pairwise comparisons for LLM quality across all four conditions.

Table 5: One-way ANOVA results for LLM quality by perceived race, per condition. The ANOVA p -value applies to the three-group test across all racial groups within each condition. Pairwise p -values are raw (unadjusted) Welch-test values (`equal_var=False`); Bonferroni significance threshold is $p < 0.0167$ for three race-pair comparisons. An asterisk (*) marks significance at this threshold. All three pairwise comparisons are shown per condition.

Condition	Comparison	ANOVA p	Raw Pairwise p	Cohen’s d
BASELINE	White vs. Black	0.040*	0.705	+0.040
	White vs. Asian		0.042	−0.215
	Asian vs. Black		0.016*	+0.255
PERSONA-BLIND	White vs. Black	0.492	0.654	−0.047
	White vs. Asian		0.476	+0.075
	Asian vs. Black		0.254	−0.121
RERANKED	White vs. Black	0.322	0.469	+0.078
	White vs. Asian		0.430	−0.085
	Asian vs. Black		0.132	+0.164
SYSTEM-PROMPT	White vs. Black	0.187	0.345	+0.100
	White vs. Asian		0.377	−0.093
	Asian vs. Black		0.068	+0.193

The BASELINE ANOVA for LLM quality reaches statistical significance ($p = 0.040$). The White vs. Asian pairwise comparison shows a medium effect size (Cohen’s $d = -0.215$, raw $p = 0.042$), indicating White-named prompts scored lower on average than Asian-named prompts in the baseline condition; this raw p -value does not survive Bonferroni correction (threshold $p < 0.0167$). The most statistically robust pairwise finding is the Asian vs. Black comparison in the baseline condition (raw $p = 0.016$, Cohen’s $d = 0.255$), which is the only comparison that survives Bonferroni correction and represents the largest effect size observed across all conditions and comparisons. After applying PERSONA-BLIND prompting, the ANOVA is no longer significant ($p = 0.492$), with all pairwise effect sizes shrinking substantially. RERANKED generation similarly yields a non-significant ANOVA ($p = 0.322$).

For length and readability, none of the four conditions yield statistically significant ANOVAs, suggesting that overt quality differences are the most sensitive marker of demographic conditioning in this dataset. Because responses sharing the same template within a cell are not fully independent (see Section 3.1), ANOVA results should be interpreted as approximate due to template-level clustering.

The baseline ANOVA significance is driven primarily by the Asian vs. Black contrast. A post-hoc test restricted to White vs. Black alone yields $p = 0.705$, which is not significant. The overall ANOVA should therefore be interpreted as reflecting a specific name-group contrast rather than a uniform quality disadvantage across all three groups.

The Asian > White > Black quality ordering warrants some discussion. One plausible account is differential name frequency in LLM training corpora: names like Arjun, Hiroshi, and Wei may appear more often in high-quality professional or academic text (e.g., technical papers, international

conference proceedings) than names like Jamal or Lakisha, which appear more frequently in audit study contexts or lower-register web text. If so, the model’s quality associations track corpus register rather than explicit racial stereotypes. This is speculative – we did not measure training-data name frequency – but it is testable in future work and would, if confirmed, shift intervention design away from name removal toward corpus debiasing at the source.

5.4 Quality by Race Across Conditions

Figure 3 shows mean LLM-as-judge quality scores by perceived race across all four conditions.

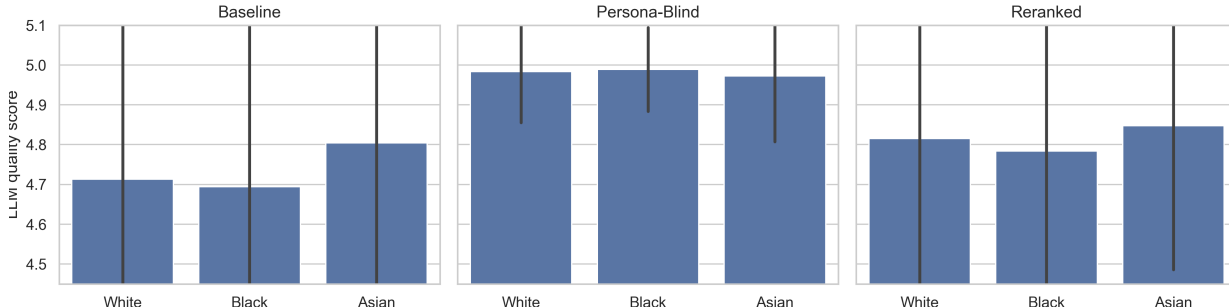


Figure 3: Mean LLM-as-judge quality score by perceived race across the three main conditions. Error bars show standard deviation. The BASELINE panel shows visible bar-height variation with Black-named prompts scoring lowest (4.694) and Asian highest (4.805). Bars are nearly level under PERSONA-BLIND, which also achieves the highest overall quality (≈ 4.98). Note: y-axis starts at 4.5 to make mean differences visible; all scores are on the original 1–5 scale. SYSTEM-PROMPT group means are reported separately in Table 8.

Under BASELINE, Black-named prompts score lowest (4.694) and Asian-named prompts score highest (4.805), with White falling between (4.713). This gap nearly vanishes under PERSONA-BLIND prompting, where all three groups cluster tightly around 4.98. The RERANKED condition shows residual within-group variation but a smaller spread than BASELINE, consistent with the intermediate parity gap in Table 3.

5.5 Task-Level Analysis

Table 6 breaks down the maximum pairwise LLM quality gap by task type for each condition.

Table 6: Maximum pairwise LLM quality parity gap by task type and condition. Values represent the largest racial group difference within each task.

Task Type	BASLINE	PERSONA-BLIND	RERANKED	SYSTEM-PROMPT
Cover Letter	0.269	0.083	0.194	0.306
Concept Explanation	0.164	0.000	0.115	0.056
Advice Giving	0.065	0.000	0.000	0.083
Problem Solving	0.031	0.000	0.000	0.028
Recommendation Letter	0.013	0.000	0.000	0.028

Demographic quality disparities are not distributed uniformly across tasks. In the BASELINE condition, **cover letters** show the largest gap (0.269), followed by concept explanations (0.164). Advice giving, problem solving, and recommendation letters show minimal disparities (≤ 0.065).

Table 7 reports one-way ANOVA results for LLM quality by perceived race within each task type, computed on the baseline condition.

Table 7: Per-task one-way ANOVA results for LLM quality by perceived race (baseline condition only). Bonferroni threshold for 5 tasks: $p < 0.010$. n varies by task due to prompt suite structure; cover letters have $n = 159$ while other tasks have $n = 94$ – 99 , so the larger sample partially explains the higher ANOVA sensitivity for that task.

Task Type	n	ANOVA p	Significant?
Cover Letter	159	0.00157	Yes*
Concept Explanation	94	0.179	No
Advice Giving	94	0.370	No
Problem Solving	99	0.355	No
Recommendation Letter	94	0.980	No

Cover letters are the only task with a statistically significant race-linked quality gap after Bonferroni correction ($p = 0.00157 < 0.010$), confirming that the global baseline ANOVA result (Table 5) is driven almost entirely by this single task type. All other tasks are non-significant by a wide margin, with recommendation letters showing essentially no race signal ($p = 0.980$). Note that cover letters have more responses ($n = 159$) than other tasks ($n = 94$ – 99); the larger sample increases statistical power for that task, which partially contributes to its being the only significant result.

Cover letters are professional identity documents where a person’s name is a central, repeated element; the model draws more strongly on stereotypical associations when generating a detailed professional narrative anchored to a specific individual. Recommendation letters use names similarly but follow tighter formulaic conventions that constrain output variation. Concept explanations, where the name functions as an addressee label rather than the document subject, show intermediate disparities.

Under PERSONA-BLIND, gaps drop sharply. Cover letters retain a residual gap (0.083) – based on a few extra 4-rated outputs in the Asian-labeled subset (5 fours vs. 2 for Black and 3 for White) – likely reflecting prompt-template sampling variation or judge noise rather than demographic conditioning, since names are not visible to the model. Four of the five task types reach an exact gap of 0.000.

RERANKED generation reduces the cover-letter gap to 0.194 (vs. 0.269 baseline) but falls well short of PERSONA-BLIND. The SYSTEM-PROMPT cover-letter gap (0.306) is larger than baseline: score distributions show Asian-associated prompts received 22 of 36 fives versus 11 of 36 for Black-associated and 13 of 36 for White-associated, meaning the name cue remained active for cover letters despite the instruction. System prompting reduces the overall race gap but does not uniformly reduce task-level gaps, and reranking’s limited effectiveness reinforces that bias originates in the input – when the name is present in every candidate, selecting among biased outputs cannot remove the cause.

Figure 4 shows the full task-by-metric parity gap heatmaps.

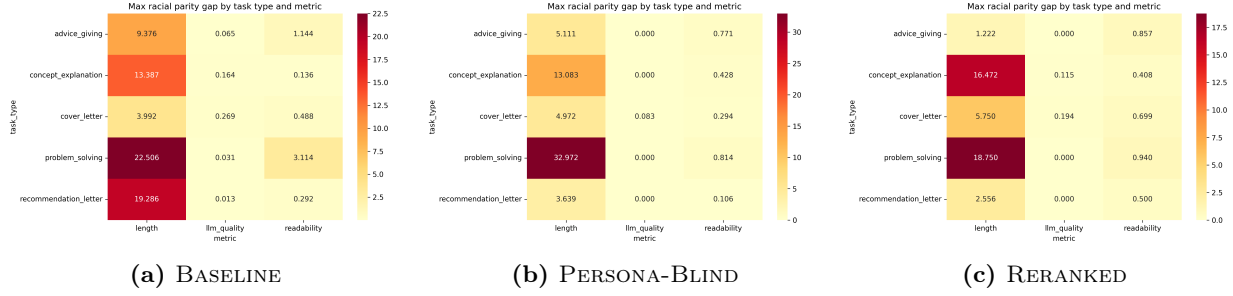


Figure 4: Task-level parity gap heatmaps across metrics (length, LLM quality, readability) and conditions. Darker cells indicate larger gaps. The BASELINE cover-letter row is the clearest high-disparity cell for LLM quality; PERSONA-BLIND substantially reduces values across the board. Readability gaps persist at moderate levels even under PERSONA-BLIND (cover-letter readability gap: 0.294), pointing to surface-feature conditioning not fully eliminated by name removal alone – a distinction that quality-only audits would miss. SYSTEM-PROMPT heatmap not shown; task-level gaps for that condition are reported in Table 6.

Length gaps are considerably larger than quality or readability gaps within most tasks across all conditions, reflecting high natural variance in response verbosity. The BASELINE cover-letter row is the dominant high-disparity cell for LLM quality, while the PERSONA-BLIND heatmap shows substantially lower values throughout.

5.6 System-Prompt Intervention Results

Table 8 summarizes results for the system-prompt condition, evaluated on the full 540-prompt suite.

Table 8: System-prompt condition: mean LLM quality by group and overall, with maximum parity gap and ANOVA result.

Condition	Asian	Black	White	Overall	Max Gap	ANOVA p
SYSTEM-PROMPT	4.867	4.794	4.833	4.831	0.072	0.187
BASELINE (ref.)	4.805	4.694	4.713	4.737	0.110	0.040*
PERSONA-BLIND (ref.)	4.972	4.989	4.983	4.981	0.017	0.492

The system-prompt condition achieves a maximum parity gap of 0.072, representing partial mitigation relative to baseline (0.110) but substantially weaker than persona-blind (0.017). The ANOVA is non-significant ($p = 0.187$), though the gap itself remains larger than persona-blind, indicating reduced but not eliminated demographic disparity.

Mitigation is assessed using both statistical significance and gap magnitude. ANOVA non-significance means the remaining group differences are not statistically detectable in this sample, while the max parity gap describes the remaining practical disparity. PERSONA-BLIND is the strongest mitigation because it achieves both a non-significant ANOVA and the smallest remaining gap (0.017). SYSTEM-PROMPT is also non-significant, but its remaining gap (0.072) is larger, so we call it partial mitigation rather than the best mitigation.

6 Discussion

6.1 Bias Appears Largely Driven by Input Cues

Removing the name from the prompt – nothing else – substantially eliminates quality gaps across all task types. The name cue is likely a primary mechanism by which disparities enter the output, though we cannot rule out that model-internal representations also contribute independently; the system-prompt condition’s partial effectiveness shows the name is not the only active factor. Our evidence says removing the name is sufficient to reduce measured disparities, not that it is the exclusive cause.

Under the Llama judge, PERSONA-BLIND raises mean quality above BASELINE (4.981 vs. 4.737); this did not replicate under Mistral (Section 6.6), so the quality increase should be treated as judge-sensitive.

6.2 Why Reranking Is Less Effective

Reranking was included as a deliberately simple output-level baseline to answer a specific question: can selecting among multiple candidates after generation reduce demographic gaps, without removing the name from the prompt? Its underperformance is a result, not a design failure – it demonstrates precisely why fixing bias at the input matters when every candidate in the pool has already been conditioned on the same demographic signal.

Because every candidate was generated from the same name-containing prompt, reranking can only select within an already-conditioned pool. Our reranking criterion – selecting the candidate closest to the per-task mean quality established by baseline outputs – anchors to a biased reference point and provides no mechanism to distinguish between responses that score similarly because they are equivalently good versus responses that score similarly because conditioning produced the same biased output repeatedly. When multiple candidates are equidistant from the target, the lowest-index candidate is selected deterministically, ensuring reproducibility. A fairness-aware reranker would need to generate counterfactual candidates across name swaps and penalize demographic sensitivity directly; that is a meaningful future direction, but it is not what this project tested.

6.3 Task Structure Modulates Disparity

Cover letters require a coherent professional identity narrative centered on a named individual; that structure gives demographic associations more surface area to influence the output. Recommendation letters use names similarly but follow tight genre conventions (endorsement, examples, closing) that leave less room for variation. The implication is practical: task structure predicts where auditing effort is most needed, and global parity gaps can hide task-level concentrations that matter for deployment decisions.

6.4 Limitations

The name lists are drawn from North American English-language audit studies and will not generalize to other cultural contexts.

The most important validity threat is that the same model generates and judges responses. We treat Llama-as-judge as a consistent scoring rubric across conditions, not a proxy for human quality. Two features of our design limit this threat: (1) parity gaps shrinking when names are

removed is hard to attribute to self-judge artifact – self-consistency bias would predict uniform or idiosyncratic shifts, not gaps that track name presence; (2) our Mistral rescoring (Section 6.6) showed the same direction of gap reduction, and non-LLM metrics (length, readability, LABE agency proxy) corroborate the quality story. Full-corpus independent rescoring remains future work. The judge scores are a consistent rubric under which the main effect holds, not ground truth.

We test only three racial groups. We use the Bertrand and Mullainathan name lists for comparability with a standard audit tradition, accepting reduced cultural breadth. Our results are a replicable slice of North American name-based perceived-race bias, not a comprehensive fairness audit. A supplementary gender check on cover letters found no statistically significant gender quality gaps under the Llama judge across all four conditions (baseline: $p = 0.757$; persona-blind: $p = 0.550$; reranked: $p = 0.928$; system-prompt: $p = 0.825$), suggesting race-associated name effects dominated. Expanding to intersectional cells, more groups, non-US name lists, and human evaluation are natural next steps.

Of 540 reranked responses, 26 returned non-numeric judge scores and were excluded (`dropna()`), leaving 514 valid entries. Missing entries concentrated in problem-solving (16) and advice-giving (7), with roughly uniform racial distribution (Asian: 10, Black: 9, White: 7). The reported gap of 0.063 was verified by manual recomputation; filling nulls with zero gives 0.083.

With 11 Asian, 12 Black, and 14 White names, we reduce reliance on any single name but cannot fully separate individual name effects from group-level effects. Within-race name variance (Asian: 0.0079, Black: 0.0184, White: 0.0072; computed as sample variance of per-name mean quality scores) exceeded between-race variance (0.0035) in the baseline condition. The within-to-between ratio quantifies this directly: within-race variance exceeds between-race variance by $2.27\times$ for Asian names, $5.29\times$ for Black names, and $2.06\times$ for White names, meaning name-level variation dominates group-level variation in all three groups. Results should therefore be read as a name-based perceived-race audit rather than a clean estimate of racial group bias. Under persona-blind prompting, within-name variance drops sharply (Asian: 0.0009, Black: 0.0006, White: 0.0010), and the within-to-between ratios fall to approximately $0.26\times$, $0.17\times$, and $0.14\times$ respectively – consistent with names, not group membership, being the primary driver of variation.

Persona-blind may also not be appropriate for all applications: some tasks need a name for coherent output.

6.5 Implementation Notes

API rate limits. Groq rate limits required request pacing during reranking ($k = 3\text{--}5$ at $T = 0.7$). We used exponential backoff and per-minute throttling. Despite this, 26 prompts returned malformed judge scores; these were excluded, with roughly uniform racial distribution across dropouts.

Length gaps increased under persona-blind. PERSONA-BLIND and RERANKED produced larger response length gaps than BASELINE, even as quality gaps shrank. Name-conditioned prompts seem to anchor the model to a specific response structure; without a name, output length varies more freely across tasks. Length parity and quality parity are partially decoupled – length-only audits can give a misleading picture.

Recommendation letters vs. cover letters. Recommendation letters showed almost no quality disparity (gap = 0.013) despite using names similarly to cover letters. Recommendation letters follow tighter genre conventions (endorsement, examples, closing) that leave less room for demo-

graphic associations to affect output. Cover letters require a more personalized narrative, which appears more susceptible.

Reproducibility. All responses were stored as JSON with `model`, `k`, `prompt_id`, and `perceived_race` fields. This let us verify post-hoc that all 540 baseline responses used `llama-3.1-8b-instant` and that no fallback triggered. Future audits should store intermediate candidate pools, not only selected responses.

6.6 Independent Judge Robustness Check

We rescored a balanced 180-response subset (60 per condition, 20 Asian / 20 Black / 20 White) using Mistral via Ollama as an independent judge – a different model family from the Groq-hosted Llama used in the main pipeline.

Mistral results are judge-sensitivity analysis, not a replication. Baseline mean quality was 4.15, reranked 4.22, and persona-blind 4.05 – the persona-blind quality increase observed under Llama did not replicate. Mistral used a coarser score distribution concentrated around 4 and 5, which makes small mean differences hard to read; the direction of parity-gap reduction was partly consistent with the main results, but the exact quality improvement is judge-dependent.

Table 9: Mistral independent-judge robustness check results (180-response subset).

Condition	Mistral Mean Quality	Mistral Max Gap
BASELINE	4.15	0.150
PERSONA-BLIND	4.05	0.100
RERANKED	4.22	0.150

The Mistral check does not replicate the persona-blind average-quality increase observed under the main Llama judge, so the average-quality improvement should be treated as judge-sensitive. This weakens claims about exact average-quality improvement, but not the main evidence that the primary Llama-based audit found a statistically significant baseline quality disparity reduced by persona-blind prompting. We treat Mistral as a sensitivity check, not a direct replication.

The score distributions from the 180-response check confirm the coarser Mistral scoring: baseline received 9 fives and 51 fours, persona-blind received 3 fives and 57 fours, and reranked received 13 fives and 47 fours; no condition received any score below 4. This distribution makes small mean differences hard to interpret, but it also means persona-blind’s lower mean under Mistral may reflect a few fewer 5-rated outputs rather than a genuine quality decline. We treat this as evidence that the exact quality improvement is judge-sensitive, not as evidence that persona-blind degrades quality.

6.7 Published-Metric Robustness Comparisons

To connect DemoGen to recent long-form generation-bias benchmarks, we ran lightweight checks inspired by LABE [7] and SAGED [5] on our saved outputs.

The LABE-inspired lexical agency proxy counted agentic and communal terms and computed agency balance per 1,000 words. The agency balance is measured as agentic word rate minus communal word rate per 1,000 words, so a baseline gap of 0.588 means the largest race-pair difference was about 0.588 net agentic terms per 1,000 words – less than one additional term, which is small in

practical terms and statistically non-significant (ANOVA $p = 0.925$). Unlike the main LLM-quality metric, the lexical agency proxy did not show evidence of race-linked disparity in our outputs.

A SAGED-inspired disparity summary matched the main result: baseline `llm_quality` maximum race gap was 0.110, persona-blind reduced it to 0.017, reranked to 0.063, and system prompt to 0.072.

At the task level, the agency proxy shows a different ordering than the quality gap: cover letters show the largest agency gap (2.92 net agentic terms per 1,000 words) and recommendation letters the smallest (1.01), but neither reaches significance. The task ordering for agency does not match the quality gap ordering, reinforcing that the two metrics capture partially distinct phenomena.

These are not full reproductions of LABE or SAGED – they apply published-paper-style diagnostics to our outputs as a lightweight consistency check.

The agency result is worth comparing directly to prior benchmarks. [7] report Llama3 reference-letter agency bias scores of 26.29 – substantially higher than ChatGPT (8.02) and Mistral (7.90) on the same task. Our proxy found no significant race-linked agency disparity ($p = 0.925$, max gap 0.588 net agentic terms per 1,000 words). Two explanations are plausible: Llama 3.1 8B may exhibit less agency bias than the earlier Llama3, or our task mix – which weights cover letters and advice-giving more heavily than reference letters – may produce less agency-salient outputs. Either way, the disconnect between a significant quality gap ($p = 0.040$) and a non-significant agency gap ($p = 0.925$) shows that quality-parity audits and agency-style audits capture different phenomena and are not redundant.

6.8 Real-World Implications

LLMs are already being deployed in hiring tools, college application assistants, and professional writing platforms where applicant names appear naturally in prompts. Our findings suggest that in these deployments, the model may be producing systematically different quality outputs depending on the perceived race associated with the applicant’s name, without the developer or user being aware of it.

The practical implication is direct: before deploying an LLM for any task where a person’s name appears in the prompt and output quality matters, developers should audit for demographic parity gaps using a methodology like DemoGen. Where gaps are found, persona-blind preprocessing – replacing names with neutral placeholders before generation – is a zero-cost intervention that our results suggest can eliminate most of the quality disparity while potentially improving average output quality. This requires no model retraining, no access to model weights, and no additional inference calls. The fact that a simple system-prompt instruction to “ignore demographic information” produced weaker results than name removal confirms that this is not a problem developers can instruct their way out of: it has to be addressed at the input level.

7 Conclusion

DemoGen audits and mitigates demographic bias in LLM-generated content across five professional writing tasks and three perceived racial groups. Baseline outputs show statistically significant quality disparities (ANOVA $p = 0.040$, max gap = 0.110), driven by Asian-associated names scoring higher than Black-associated names ($p = 0.016$, Cohen’s $d = 0.255$). The White vs. Black contrast was not significant ($p = 0.705$); the effect is not a uniform disadvantage across all non-White groups.

Persona-blind prompting reduces the maximum quality gap by 85% (to 0.017) and raises average quality from 4.737 to 4.981 under the Llama judge; the quality increase did not replicate under Mistral. Reranking gives partial improvement (gap = 0.063, mean = 4.815) but is structurally constrained – all candidates are drawn from already-biased prompts. The system-prompt instruction reduced the quality gap to 0.072 but left the readability gap nearly unchanged (0.771 vs. 0.767 baseline), showing that explicit instruction is not enough when conditioning operates at the input level. Cover letters show the highest disparity (baseline gap = 0.269); advice-giving and recommendation-letter tasks show minimal gaps.

Global parity gaps hide task-level concentrations. Persona-blind preprocessing requires no re-training, no model access, and no extra inference calls, yet cuts the quality gap by 85%. A system-prompt instruction cannot match this – the problem has to be fixed at the input.

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